WORKED SOLUTION

STEP III 2011, Question 6

Pure • Integral Substitutions

STEP III 2011, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q6

The key idea

We are not allowed to evaluate anything, so the only tools are the ones that turn one integral into another: substitution and integration by parts. The limits are the clue. The numbers $\frac{1}{3}, \frac{1}{2}$ and $\ln 2, \ln 3$ are linked by $v = e^{-u}$, and $\frac{1}{2}\ln 2, \frac{1}{2}\ln 3$ are linked to $\frac{1}{2}, \frac{1}{3}$ by $t = e^{-2x}$. The integrals T and V have the same limits, and $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{artanh} t = \frac{1}{1-t^2}$ suggests integration by parts between them. Three links are enough to make all four integrals equal.



Solution

Throughout we use

$$\operatorname{artanh} t = \tfrac{1}{2}\ln\frac{1+t}{1-t} \quad (|t|<1), \qquad \frac{\mathrm{d}}{\mathrm{d}t}\operatorname{artanh} t = \frac{1}{1-t^2},$$

and in particular $\operatorname{artanh}\frac{1}{2} = \frac{1}{2}\ln 3$ and $\operatorname{artanh}\frac{1}{3} = \frac{1}{2}\ln 2$. All the integrands are continuous on their (closed, bounded) ranges, so every integral exists and the manipulations below are legitimate.

$T = V$: integration by parts

In T we integrate $\frac{1}{t}$ and differentiate $\operatorname{artanh} t$, taking $u = \operatorname{artanh} t$ and $\mathrm{d}w = \frac{\mathrm{d}t}{t}$, so $w = \ln t$ (fine, as $t > 0$ on the range):

$$\begin{aligned} T &= \Big[\ln t \operatorname{artanh} t\Big]_{1/3}^{1/2} - \int_{1/3}^{1/2} \frac{\ln t}{1-t^2}\,\mathrm{d}t \\ &= \Big[\ln t \operatorname{artanh} t\Big]_{1/3}^{1/2} + V, \end{aligned}$$

since the remaining integral is exactly V (the name of the dummy variable doesn't matter).

> **Ask yourself** Doesn't the boundary term spoil the equality?

It vanishes, and this is the neat part of the question. At the upper limit $\ln\frac{1}{2}\cdot\operatorname{artanh}\frac{1}{2} = (-\ln 2)(\frac{1}{2}\ln 3)$; at the lower limit $\ln\frac{1}{3}\cdot\operatorname{artanh}\frac{1}{3} = (-\ln 3)(\frac{1}{2}\ln 2)$. Both are $-\frac{1}{2}\ln 2\ln 3$, so

$$\Big[\ln t\,\operatorname{artanh} t\Big]_{1/3}^{1/2} = -\tfrac{1}{2}\ln 2\ln 3 + \tfrac{1}{2}\ln 2\ln 3 = 0,$$

and therefore $T = V$.

$V = U$: the substitution $v = e^{-u}$

Put $v = e^{-u}$, so $\ln v = -u$ and $\mathrm{d}v = -e^{-u}\,\mathrm{d}u$. When $v = \frac{1}{3}$, $u = \ln 3$; when $v = \frac{1}{2}$, $u = \ln 2$. Then

$$\begin{aligned} V &= -\int_{\ln 3}^{\ln 2} \frac{-u}{1-e^{-2u}}\left(-e^{-u}\right)\mathrm{d}u = -\int_{\ln 3}^{\ln 2} \frac{u\,e^{-u}}{1-e^{-2u}}\,\mathrm{d}u \\ &= \int_{\ln 2}^{\ln 3} \frac{u\,e^{-u}}{1-e^{-2u}}\,\mathrm{d}u \qquad \text{(swapping the limits)} \\ &= \int_{\ln 2}^{\ln 3} \frac{u}{e^{u}-e^{-u}}\,\mathrm{d}u \qquad \text{(multiplying top and bottom by } e^{u}\text{)} \\ &= \int_{\ln 2}^{\ln 3} \frac{u}{2\sinh u}\,\mathrm{d}u = U. \end{aligned}$$

$X = T$: the substitution $t = e^{-2x}$

First rewrite the integrand of X. For $x > 0$,

$$\coth x = \frac{e^{x}+e^{-x}}{e^{x}-e^{-x}} = \frac{1+e^{-2x}}{1-e^{-2x}}, \qquad \text{so} \qquad \ln(\coth x) = \ln\frac{1+e^{-2x}}{1-e^{-2x}} = 2\operatorname{artanh}\left(e^{-2x}\right),$$

which is valid because $0 < e^{-2x} < 1$. Now put $t = e^{-2x}$, so $\mathrm{d}t = -2t\,\mathrm{d}x$, that is $\mathrm{d}x = -\frac{\mathrm{d}t}{2t}$. When $x = \frac{1}{2}\ln 2$, $t = \frac{1}{2}$; when $x = \frac{1}{2}\ln 3$, $t = \frac{1}{3}$. Hence

$$X = \int_{1/2}^{1/3} 2\operatorname{artanh} t\left(-\frac{1}{2t}\right)\mathrm{d}t = \int_{1/3}^{1/2} \frac{\operatorname{artanh} t}{t}\,\mathrm{d}t = T.$$

Conclusion

Putting the three links together, $X = T = V = U$, so all four integrals are equal, as required. We never had to find a value for any of them.

Checks

Simpson's rule on each integral separately gives $T \approx U \approx V \approx X \approx 0.177704$, agreeing to about 12 decimal places. A quick check of the vanishing boundary term on

a calculator is also reassuring: $\ln\frac{1}{2}\cdot\operatorname{artanh}\frac{1}{2}$ and $\ln\frac{1}{3}\cdot\operatorname{artanh}\frac{1}{3}$ both come out as $-0.3807\ldots$.

Common slips

- Losing a minus sign in $V = U$: there are three of them (the one in front of V, $\ln v = -u$ and $\mathrm{d}v = -e^{-u}\,\mathrm{d}u$), and swapping the limits supplies a fourth.
- Abandoning the integration by parts because the boundary term looks non-zero, without evaluating it at both ends.
- Writing $\ln\coth x = \operatorname{artanh}(e^{-2x})$ and dropping the factor 2; it is cancelled later by the $\frac{1}{2}$ in $\mathrm{d}x = -\frac{\mathrm{d}t}{2t}$.

Beyond STEP

Expanding $\operatorname{artanh} t/t$ as a power series shows that the common value is $\chi_2(\frac{1}{2}) - \chi_2(\frac{1}{3})$, where $\chi_2(z) = \sum_{k\geq 0} z^{2k+1}/(2k+1)^2$ is Legendre's chi function, a close relative of the dilogarithm $\mathrm{Li}_2(z) = \sum_{n\geq 1} z^n/n^2$. Many identities for these functions come from exactly this kind of change of variable. Showing that two integrals agree without computing either is a standard technique in analysis, and the theorems that justify substitution and integration by parts (continuity, the fundamental theorem of calculus) are made precise in a first analysis course. Abbott's *Understanding Analysis* is a good place to see them proved carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.